Variable separation method.

SOLUTIONS TO EXAMPLES

1. Laplace's equations $\nabla^2 u = 0$ in plane polar coordinates reads

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u}{\partial \phi^2} = 0. \tag{1}$$

We seek its solution in the form

$$u(r,\phi) = R(r)\Phi(\phi). \tag{2}$$

Substituting this into Eq. (1) gives

$$\Phi(\phi)\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial R}{\partial r}\right) + R(r)\frac{1}{r^2}\frac{\partial^2 \Phi}{\partial \phi^2} = 0, \tag{3}$$

where we can also change the partial derivatives (∂) to the total derivatives (d), since each of R and Φ depend on one variable only (r and ϕ, respectively). Dividing Eq. (3) through by $R(r)\Phi(\phi)$ gives

$$\underbrace{\frac{1}{R(r)}\frac{1}{r}\frac{d}{dr}\left(r\frac{dR}{dr}\right)}_{\text{depends on } r \text{ only}} + \frac{1}{r^2}\underbrace{\frac{1}{\Phi(\phi)}\frac{d^2\Phi}{d\phi^2}}_{\text{const}} = 0. \tag{4}$$

For this equation to hold for all r and ϕ, the factor that depends on ϕ must be equal to a constant. Denoting this separation constant by λ gives the equations for $\Phi(\phi)$ and $R(r)$:

$$\frac{1}{\Phi(\phi)}\Phi''(\phi) = \lambda \quad \Leftrightarrow \quad \Phi''(\phi) = \lambda\Phi(\phi), \tag{5}$$

and

$$\frac{1}{R(r)}\frac{1}{r}(rR')' + \frac{\lambda}{r^2} = 0 \quad \Leftrightarrow \quad \frac{1}{r}(rR')' + \frac{\lambda}{r^2}R(r) = 0. \tag{6}$$

As we will see, λ cannot be positive.

<u>For $\lambda = 0$</u>, Eq. (5) gives

$$\Phi''(\phi) = 0.$$

Integrating this equation gives $\Phi'(\phi) = C_1$, and integrating again yields

$$\Phi(\phi) = C_1\phi + C_2,$$

where C_1 and C_2 are arbitrary constants. The angular part of the solution must be a periodic function of ϕ with a period 2π, i.e.,

$$\Phi(\phi + 2\pi) = \Phi(\phi), \tag{7}$$

which gives $C_1(\phi + 2\pi) + C_2 = C_1\phi + C_2 \;\Leftrightarrow\; 2\pi C_1 = 0 \;\Leftrightarrow\; C_1 = 0$. Hence,

$$\Phi = C \quad \text{(const)}. \tag{8}$$

The radial equation (6) for $\lambda = 0$ reads

$$\frac{1}{r}(rR')' = 0,$$

or

$$(rR')' = 0.$$

Integrating this equation gives:

$$\begin{aligned}
\int \frac{d}{dr}\left(rR'\right) dr &= \int 0 dr, \\
rR' &= C_1, \\
\frac{dR}{dr} &= \frac{C_1}{r}, \\
\int \frac{dR}{dr} dr &= \int \frac{C_1}{r} dr, \\
R(r) &= C_1 \ln r + C_2.
\end{aligned} \tag{9}$$

Combining Eqs. (8) and (9), we obtain

$$\begin{aligned}
u(r, \phi) &= (C_1 \ln r + C_2)C \\
&= CC_1 \ln r + CC_2.
\end{aligned}$$

Renaming the products of arbitrary constants CC_1 by C and C_2C by D, we have

$$u(r, \phi) = C \ln r + D. \tag{10}$$

Considering $\lambda < 0$, it is convenient to set $\lambda = -k^2$, with $k > 0$. Equation (5) then reads

$$\Phi'' + k^2\Phi = 0.$$

This is a familiar 2nd-order linear differential equation with constant coefficients. Its general solution is

$$\Phi(\phi) = A_k \cos k\phi + B_k \sin k\phi,$$

where A_k and B_k are arbitrary coefficients. Periodicity, Eq. (7), requires

$$A_k \cos(k\phi + 2\pi k) + B_k \sin(k\phi + 2\pi k) = A_k \cos k\phi + B_k \sin k\phi.$$

For this to hold for any ϕ, $2\pi k$ must be an integer number of full angles. Hence, k must be integer, and since $k > 0$, we have[1]

$$k = 1,\, 2, \ldots, \quad \text{i.e.,} \quad k = n, \quad n \in \mathbb{N},$$

so that

$$\Phi(\phi) = A_n \cos n\phi + B_n \sin n\phi. \tag{11}$$

(Note that for $\lambda > 0$ we would set $\lambda = k^2$ and have $\Phi'' - k^2\Phi = 0$, with the general solution $\Phi(\phi) = Ae^{k\phi} + Be^{-k\phi}$, which is not periodic for any k.)

For $\lambda = -n^2$ the radial equation (6) gives

$$\frac{1}{r}\left(rR'\right)' - \frac{n^2}{r^2}R(r) = 0,$$

or

$$r\left(rR'\right)' = n^2R(r). \tag{12}$$

We seek solution of this 2nd-order equation in the form $R(r) = r^s$, for which

$$R' = sr^{s-1}, \qquad rR' = sr^s, \qquad r(rR')' = rs^2r^{s-1} = s^2r^s.$$

[1] There is no need to consider negative k, since $\cos(-k\phi) = \cos k\phi$ and $\sin(-k\phi) = -\sin k\phi$, and the minus sign in the latter can be absorbed into the arbitrary constant B_k.

Hence, Eq. (12) gives

$$s^2 r^s = n^2 r^s,$$

so that $s^2 = n^2$ and $s = \pm n$. Hence, the two independent solutions of the radial equation are

$$R(r) = r^n, \qquad \text{and} \qquad R(r) = r^{-n}.$$

Combining each of these with the angular part (11) gives the following solutions

$$u(r,\phi) = r^n (A_n \cos n\phi + B_n \sin n\phi), \tag{13}$$

$$u(r,\phi) = r^{-n} (\tilde{A}_n \cos n\phi + \tilde{B}_n \sin n\phi), \tag{14}$$

where A_n, B_n, $\tilde{A}_n$ and $\tilde{B}_n$ are arbitrary coefficients[2].

Since Laplace's equation $\nabla^2 u = 0$ is linear and homogeneous, we apply the superposition principle to construct the more general solution of this equation from those in Eqs. (10), (13) and (14):

$$u(r,\phi) = C \ln r + D + \sum_{n=1}^{\infty} r^n (A_n \cos n\phi + B_n \sin n\phi) + \sum_{n=1}^{\infty} r^{-n} (\tilde{A}_n \cos n\phi + \tilde{B}_n \sin n\phi). \tag{15}$$

Note that the first term and all the terms in the second sum are singular at $r = 0$. If we were asked to find the solution of the Laplace equation in a domaine that included the origin, then we would set $C = \tilde{A}_n = \tilde{B}_n = 0$.

2 (a) Here we consider the wave equation for a horizontal string acted upon by gravity,

$$u_{tt} - c^2 u_{xx} = g, \tag{16}$$

with the fixed-end boundary conditions

$$u(0,t) = 0, \quad u(l,t) = 0. \tag{17}$$

Equation (16) is inhomogeneous. We seek its solution in the form

$$u(x,t) = u_s(x) + \tilde{u}(x,t), \tag{18}$$

and choose the time-independent part $u_s(x)$ so that it satisfies (16) and (17). Hence,

$$-c^2 \frac{d^2 u_s}{dx^2} = g, \tag{19}$$

or

$$\frac{d^2 u_s}{dx^2} = -\frac{g}{c^2}.$$

Integrating, we have

$$\frac{du_s}{dx} = -\frac{g}{c^2} x + C,$$

and integrating again,

$$u_s(x) = -\frac{gx^2}{2c^2} + Cx + D.$$

Applying the boundary conditions (17), we obtain

$$u_s(0) = D = 0.$$

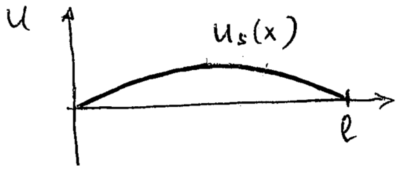
For $x = l$, $u_s(l) = 0$ gives

$$-\frac{gl^2}{2c^2} + Cl = 0 \quad \Rightarrow \quad C = \frac{gl}{2c^2}.$$

[2] Note that the terms in the solutions (13) and (14) are proportional to $\operatorname{Re} z^n$ and $\operatorname{Im} z^n$, and $\operatorname{Re} z^{-n}$ and $\operatorname{Im} z^{-n}$, respectively, where $z = re^{i\phi}$ is a complex number. This is in complete accord with the result from Complex Analysis that the real and imaginary parts of any analytical function satisfy Laplace's equation.

Therefore,

$$u_s(x) = -\frac{gx^2}{2c^2} + \frac{glx}{2c^2} = \frac{gx(l-x)}{2c^2}. \tag{20}$$



The graph of $u_s(x)$ is a parabola. This solution describes the shape of the string at equilibrium. We see that the string "sags" under gravity (the positive direction of u being in the direction of g).

Let us now substitute $u(x,t)$ from Eq. (18) into Eq. (16),

$$\tilde{u}_{tt} - c^2\tilde{u}_{xx} - c^2u_s'' = g,$$

and since u_s satisfies Eq. (19),

$$\tilde{u}_{tt} - c^2\tilde{u}_{xx} = 0.$$

The boundary conditions (17) are satisfied by $u_s(x)$, so we also have $\tilde{u}(0,t) = \tilde{u}(l,t) = 0$. Hence, $\tilde{u}(x,t)$ satisfies the homogeneous wave equation with zero boundary condition, and the solution to this problem is obtained in Sec. 2.2 of the lecture notes.

2 (b) We now consider the problem

$$u_{tt} - c^2u_{xx} = 0, \quad u(0,t) = 0, \quad u(l,t) = h. \tag{21}$$

We first find a time-independent solution $u_s(x)$ of this problem. We have

$$c^2u_s'' = 0, \quad \text{or} \quad u_s'' = 0. \tag{22}$$

Integrating this equation twice gives

$$u_s(x) = C_1x + C_2.$$

Applying the first boundary conditions from Eq. (21) to $u_s(x)$ gives $u_s(0) = C_2 = 0$. The boundary condition at $x = h$ then yields $C_1 l = h$, so that $C_1 = h/l$, and

$$u_s(x) = \frac{h}{l}x. \tag{23}$$

We now seek solution of the full problem (21) in the form

$$u(x,t) = u_s(x) + \tilde{u}(x,t).$$

Substituting this into the wave equation, we have

$$\tilde{u}_{tt} - c^2\tilde{u}_{xx} - c^2u_s'' = 0,$$

and using (22), see that

$$\tilde{u}_{tt} - c^2\tilde{u}_{xx} = 0,$$

i.e., $\tilde{u}(x,t)$ satisfies the (homogeneous) wave equation. Applying the boundary conditions from (21), and using Eq. (23), we have

$$\begin{aligned} u_s(0) + \tilde{u}(0,t) = 0 \quad &\Rightarrow \quad 0 + \tilde{u}(0,t) = 0 \quad \Rightarrow \quad \tilde{u}(0,t) = 0, \\ u_s(l) + \tilde{u}(l,t) = h \quad &\Rightarrow \quad h + \tilde{u}(l,t) = h \quad \Rightarrow \quad \tilde{u}(l,t) = 0. \end{aligned}$$

We see that $\tilde{u}(x,t)$ satisfies the homogeneous wave equation with zero boundary condition. The corresponding solution is worked in Sec. 2.2 of the lecture notes.